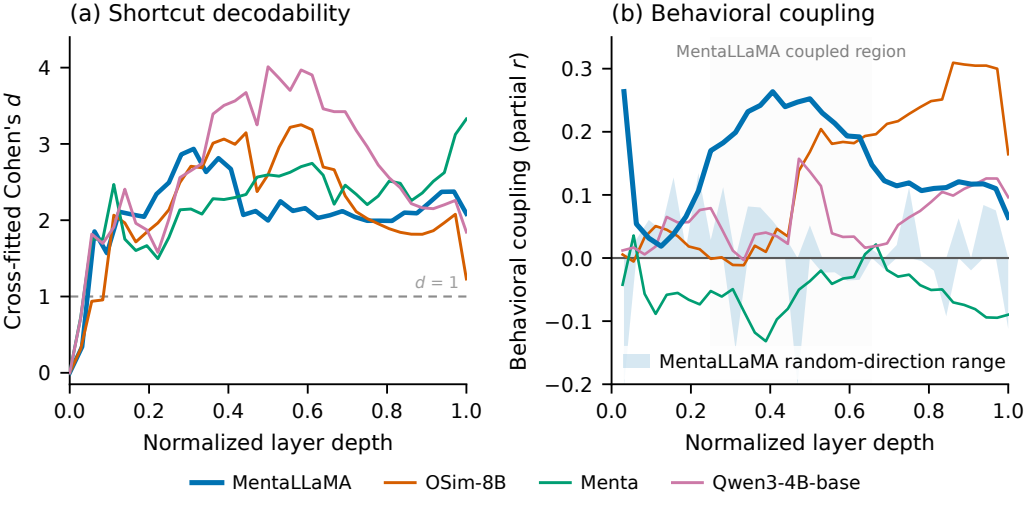
(a) Shortcut decodability
Cross-fitted Cohen's d
4
3
2
1
0
d = 1
0.0
0.2
0.4
0.6
0.8
1.0
Normalized layer depth
(b) Behavioral coupling
Behavioral coupling (partial r)
MentaLLaMA coupled region
0.3
0.2
0.1
0.0
−0.1
−0.2
MentaLLaMA random-direction range
0.0
0.2
0.4
0.6
0.8
1.0
Normalized layer depth
MentaLLaMA
OSim-8B
Menta
Qwen3-4B-base